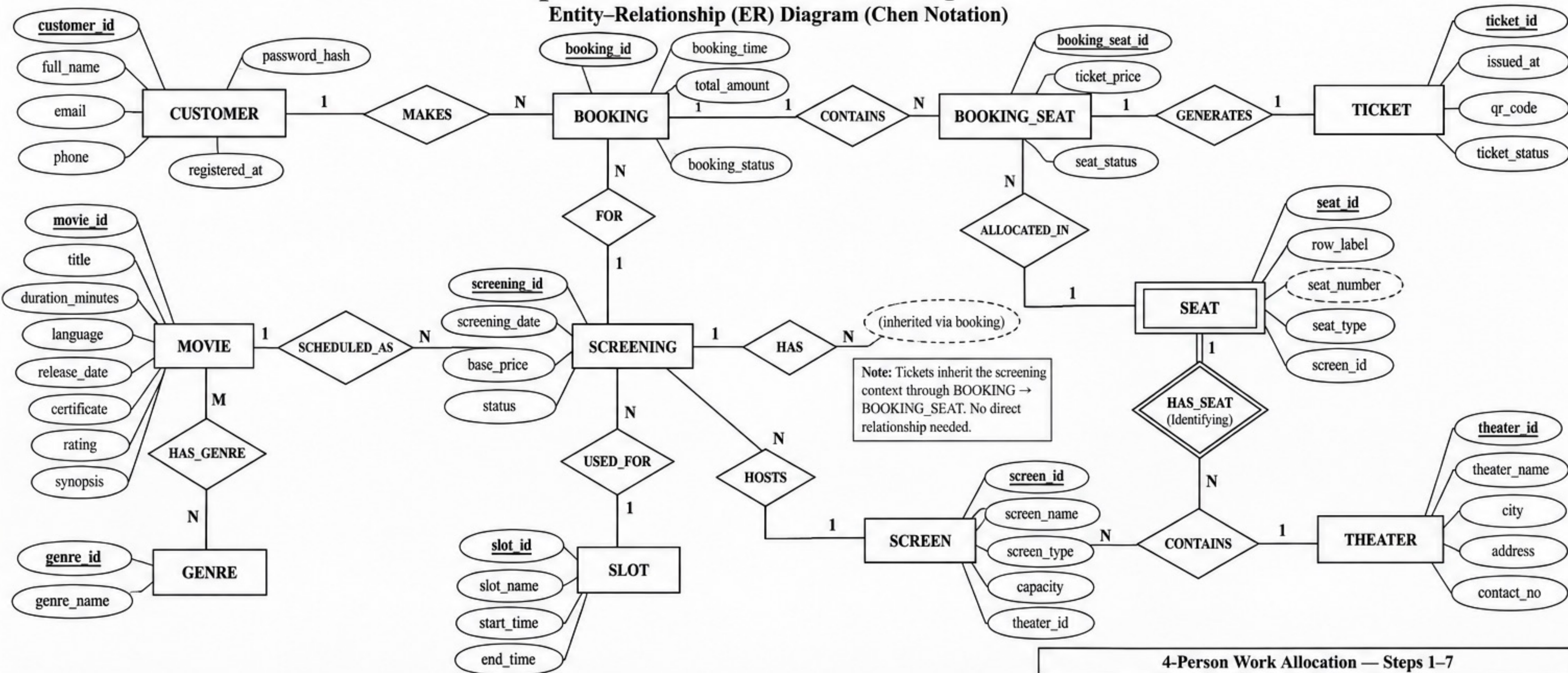


CineSphere – Movie Theater Booking Database

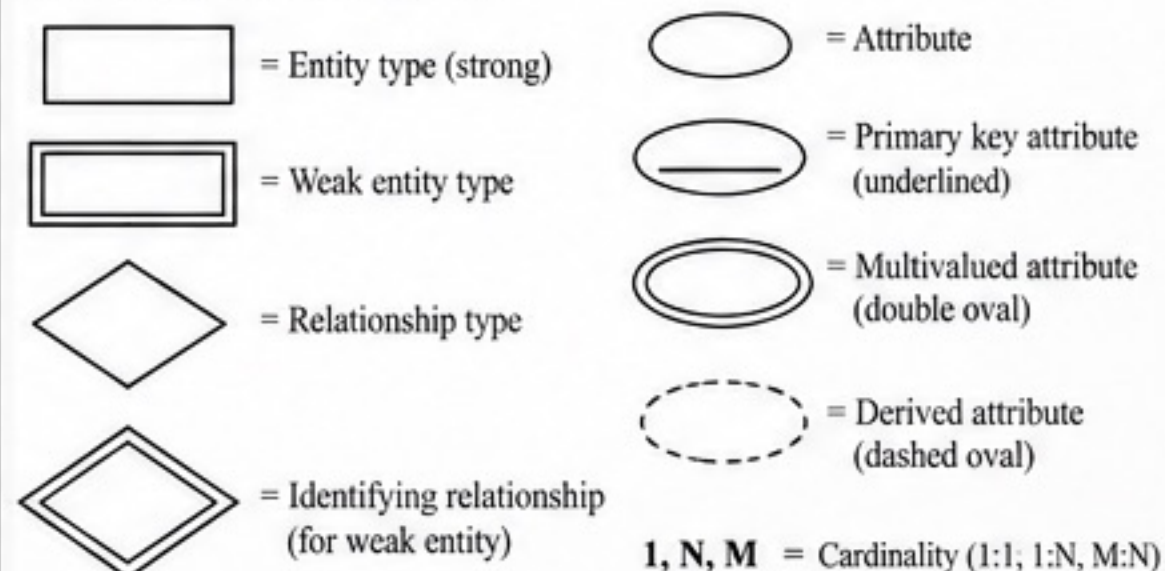
Entity–Relationship (ER) Diagram (Chen Notation)



Constraints / Business Rules

1. A screen belongs to exactly one theater.
2. A screening uses exactly one movie, one screen and one time slot.
3. A screen cannot have overlapping screenings at the same date/time.
4. A booking belongs to exactly one customer and one screening.
5. A seat can be booked only once for a particular screening.
6. Booking total equals the sum of ticket prices for its booking seats.
7. Payment amount must equal booking total for a successful payment.
8. Screening base price may be adjusted by seat type through ticket_price.
9. Seat capacity should be consistent with the number of seats defined for the screen.

Chen Notation Legend



4-Person Work Allocation — Steps 1–7

Person	Requirements & Domain Modeling
Person 1	- Step 1: Title selection - Step 2: Problem statement, actors, scope, assumptions, business rules - Own CUSTOMER, MOVIE, GENRE domain analysis and initial requirements
Person 2	ER Modeling - Step 3: Objectives - Step 4: Requirements gathering - Major ER design for THEATER, SCREEN, SEAT, SLOT, SCREENING - Define cardinalities, keys and constraints
Person 3	ER-to-Relational Design - Step 5: Complete ER modeling - Step 6: Schema design and normalization - Convert BOOKING, BOOKING_SEAT, PAYMENT, TICKET and all relationships - Document 1NF, 2NF, 3NF/BCNF decisions
Person 4	SQL Implementation - Step 7: Write DDL, DML, DCL - Create all tables, primary/foreign keys, UNIQUE/CHECK constraints - Insert realistic sample data and prepare test SELECT queries - Review another member's work (collaborative)
Shared integration	- All four members jointly verify ER ↔ schema consistency - Check normalization, referential integrity and business rules - Prepare final documentation, screenshots and presentation